ARCHITECTURE OVERVIEW

LevelFit AI is built on a **multi-layered architecture** designed for scalability, modularity, and performance optimization.

Each layer is independently deployable and communicates through well-defined API interfaces.

Frontend Layer

The frontend is responsible for delivering a seamless and interactive user experience.

Technologies Used

- React.js (component-based framework)
- Tailwind CSS (utility-first styling)

Responsibilities

- Rendering user interface
- Handling user interactions
- Displaying real-time updates
- Managing navigation between screens

Key Components

- Dashboard
- Level Map
- Workout Interface
- Diet Tracker
- Progress Analytics

Backend Layer

The backend acts as the **central processing unit** of the application.

Technologies Used

- Node.js / Appwrite Functions

Responsibilities

- User authentication and authorization
- Business logic execution
- Workout and diet data processing
- Scoring engine execution
- Level progression management

Database Layer

The database layer ensures persistent storage and efficient data retrieval.

Technologies Used

- PostgreSQL / Appwrite Database

Data Stored

- User profiles
- Workout logs
- Diet records
- Level progress
- Streak data

AI Services Layer

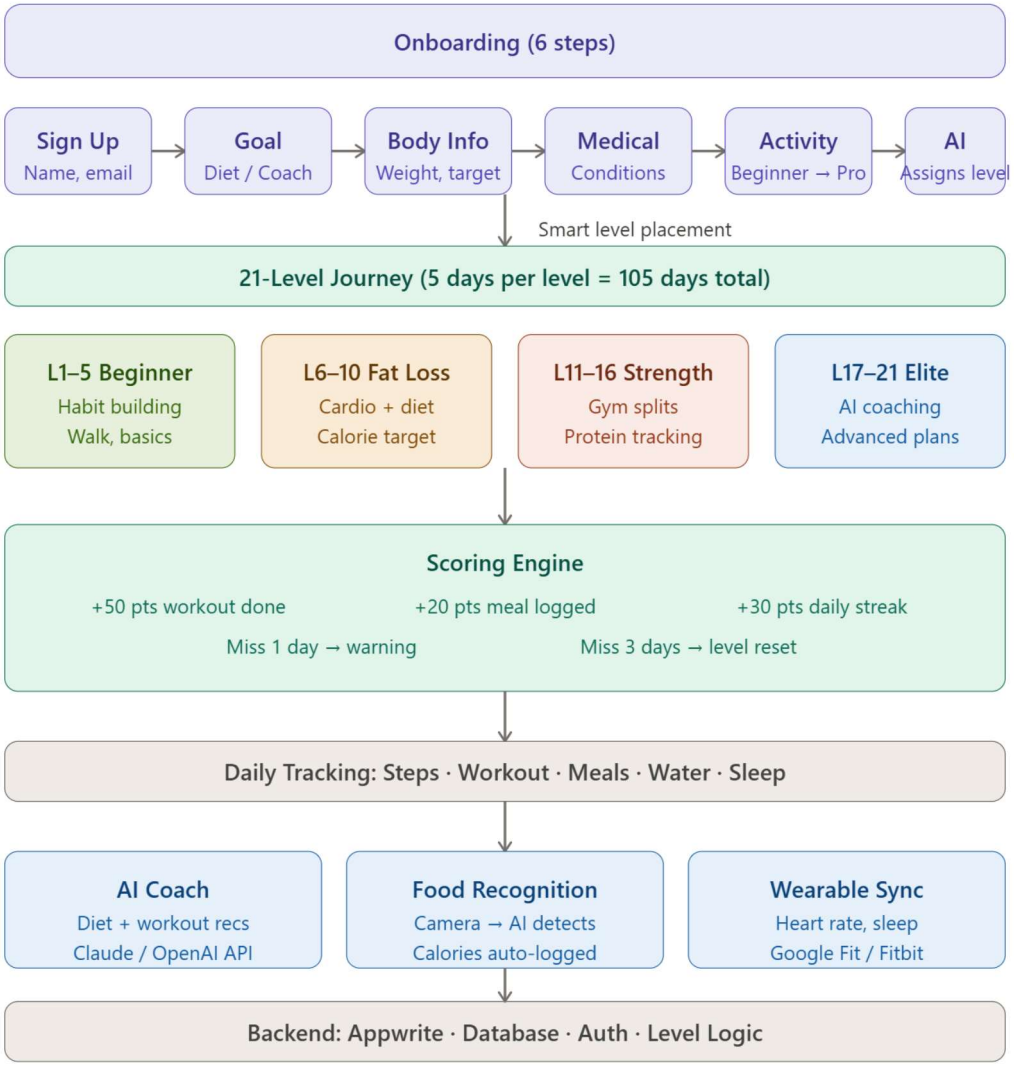
AI services enhance personalization and user engagement.

Capabilities

- Personalized workout suggestions
- Diet recommendations
- Behavioral analysis
- Virtual coaching

System Communication Flow

User → Frontend → Backend → Database → AI Services → Response

This ensures real-time synchronization across all components.

Architecture diagram